EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examinations 2018
General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME	

CIVICS GROUP	1	7	-			REGISTRATION NUMBER	

H2 Biology 9744/03

Paper 3 Long Structured and Free Response Questions

18 September 2018

2 hours

Candidates answer **Section A** on the Question Paper and **Section B** on separate writing paper.
Additional Materials: Writing Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.
Tie your answers for Section B together using the string provided.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.
Write your answer to each part of the question on a fresh sheet of paper.

The use of an approved scientific calculator is expected, where appropriate.

At the end of the examination, submit section A and B separately.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
4	
5	
Total	75

This document consists of **15** printed pages and **1** blank page.

BLANK PAGE

Section A

Answer **all** the questions in this section.

1 Human diseases can be caused by many different types of organisms, such as bacteria and viruses.

(a) State two differences between the genetic material of bacteria and viruses.

……………………………………………………………………………………………………

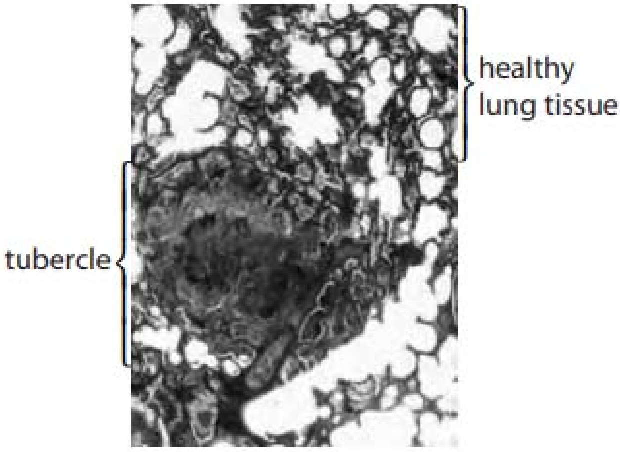
……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[2]

1. Bacteria have DNA as genome while viruses have either DNA or RNA as genome;
2. Bacteria have double stranded DNA as genetic material while viruses have either single stranded or double stranded nucleic acids as genetic material;
3. Bacteria have circular genetic material while viruses have linear genetic material;
4. Bacteria may have plasmids while viruses do not have plasmids;

Tuberculosis is caused by the bacterium *Mycobacterium tuberculosis*. Fig. 1.1 shows a tubercle in part of a lung infected by the bacteria.



©John Burbidge/Science Photo Library

Magnification ×50

Fig. 1.1

(b) Describe the role of the tubercle in transmission of tuberculosis to non-infected individuals.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[2]

1. In an infected person, tubercle cavity enlarges and tubercle ruptures, *M. tuberculosis* spill into a bronchiole and spread throughout the lungs;
2. This results in development of a productive cough that facilitates aerosol spread of infectious bacteria;

(c) Mycolic acids are substances that form part of the cell wall of *Mycobacterium tuberculosis*. Isoniazid (INH) is an antibiotic that is used to treat tuberculosis. Fig. 1.2 shows the mode of action of isoniazid. Inactive isoniazid must be activated by a bacterial enzyme, **B,** to an active form.

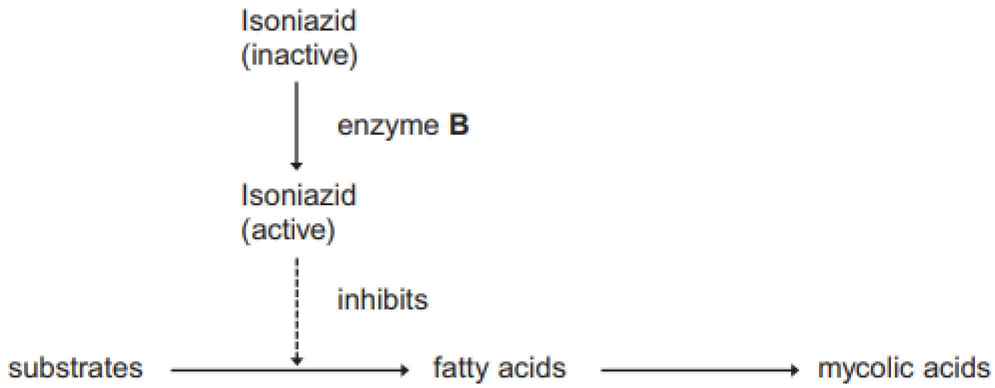


Fig. 1.2

(i) Explain the meaning of the term antibiotic.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[2]

1. Substance / chemical / molecule produced by microorganisms / fungi / bacteria / artificially produced;
2. That kills / inhibits growth of other microorganisms / bacteria / pathogens;
   R: viruses

(ii) Explain the effect of active isoniazid on *Mycobacterium tuberculosis*.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[3]

1. Presence of isoniazid prevents mycolic acids formation and hence cell wall not formed / cell wall production is inhibited / cell wall weakened;
   R: cell wall broken down
2. Water potential is lower in bacteria / water potential is higher outside of bacteria and hence water enters bacteria and turgor pressure within bacteria cells increases;
3. Resulting in osmotic lysis of bacterial cells;

(d) Treating *Mycobacterium tuberculosis* infections can be a problem, as the bacteria are resistant to many antibiotics.

(i) Explain how mutation can result in resistance to isoniazid in *Mycobacterium tuberculosis*.

....

....

....

....[2]

1. Mutated allele codes for a non-functional enzyme B;
2. Unable to catalyse the conversion of inactive isoniazid to active isoniazid / cannot catalyse formation of active form of isoniazid;

(ii) Describe one way in which genes for antibiotic resistance are transferred horizontally between bacteria.

....

....

....

....

....

....[3]

1. Via conjugation;
2. Sex pilus of donor bacterium attaches to recipient bacteria and a mating bridge is formed subsequently;
3. One strand of plasmid containing antibiotic resistance allele is nicked and transferred to recipient bacterium via mating bridge;

   OR
4. Via transformation;
5. Naked DNA containing allele for antibiotic resistance is taken up by recipient bacterium;
6. Integrated into recipient bacterium's chromosome via homologous recombination;

   OR
7. Via transduction;

8. Bacteriophage transfers the antibiotic resistance allele from donor bacterium to recipient bacterium;
9. Integrated into recipient bacterium's chromosome via homologous recombination;

(iii) The percentage of strains of *Mycobacterium tuberculosis* resistant to the antibiotic INH has increased from 2006 to 2008.

Explain how natural selection could have resulted in this increase.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………[3]

1. Antibiotic INH serves as a selection pressure;
2. Only bacteria that are resistant to INH are selected for / survive;
3. and divide / reproduce / replicate via binary fission and pass on the allele for INH resistance to daughter bacterial cells, increasing allele frequency of INH resistance;
4. the increase in number of resistant bacteria will increase the likelihood of new strains acquiring the INH resistance allele;

The ingestion of antibiotics can also have effects on the gut flora in an individual. Gut flora refers to the community of microorganisms in an individual's gastrointestinal tract. Fig. 1. 3 shows the diversity of the gut flora of an individual over the course of 18 months after taking a course of antibiotics for seven days.

The percentage of each type of bacteria in the gut flora was recorded for a period of 18 months.

Each type of bacteria is represented by a different letter in Fig. 1.3.

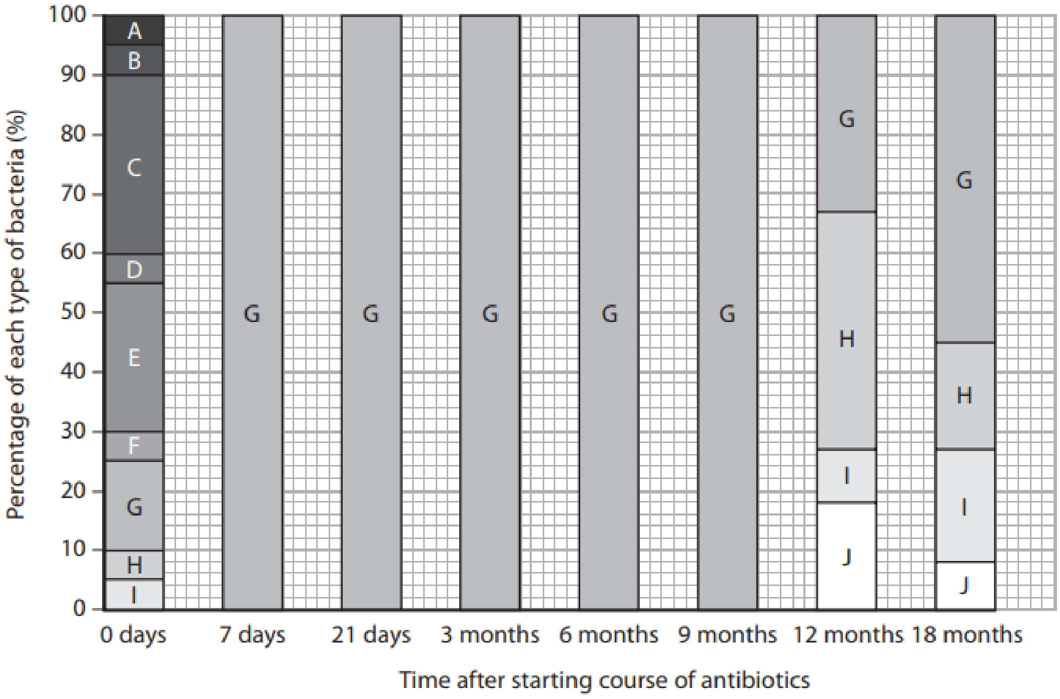


Fig. 1.3

(e) Using Fig. 1.3, describe the effect of this course of antibiotics on the diversity of gut flora in the individual.

....

....

....

....

....

....[3]

1. **Taking the antibiotic** resulted in the immediate effect of **eliminating all bacteria in the gut flora except for bacteria G**;
2. **Bacteria G** remained as the **only bacteria in the gut** from **7 days to 9 months** of the antibiotic course;
3. The variety of bacteria **increased** to **4 types of bacteria, G, H, I and J** at **12 months** after taking the antibiotic;
   OR
   The variety of bacteria **increased** to **4 types of bacteria, G, H, I and J** at **12 months to 18 months** after taking the antibiotic;
   OR
   Appearance of a **new type of bacteria, J**, at **12 months**;

(f) *Mycobacterium tuberculosis* faces various stressful conditions inside the host, such as changes in pH, exposure to reactive oxygen species and degradative enzymes. The bacterium responds to such conditions through expression of the genes in the *mymA* operon.

The seven genes in the *mymA* operon, *RV3083* to *RV3089*, are involved in the modification of fatty acids required for the cell wall of pathogenic mycobacteria, and plays an important role in their virulence.

Fig. 1.4 shows the *mymA* operon.

Promoter	Operator	*RV3083*	*RV3084*	*RV3085*	*RV3086*	*RV3087*	*RV3088*	*RV3089*

Fig. 1.4

(i) Compare *mymA* operon and *trp* operon.

....

....

....

....

....

....[3]

Similarity (max 1)

1. Repressor binds to operator to turn "off" operon;
2. Multiple structural genes under the control of the same promoter;

Differences (max 2)

3. *mymA* operon has seven structural genes while *trp* operon has five structural genes;
4. *mymA* operon is an inducible operon while trp operon is a repressible operon;
5. Gene products of the *mymA* operon are involved in the modification of fatty acids while gene products of the *trp* operon are enzymes involved in the synthesis of tryptophan;

(ii) A drop in pH results in the expression of *mymA* operon and another gene in *Mycobacteirum tuberculosis* which encodes a transcriptional regulator, VirS. VirS belongs to the AraC family and is involved in regulating the expression of the *mymA* operon.

Suggest how VirS regulates the expression of the *mymA* operon.

....

....

....

....

....

……………………………………………………………………………………………………[3]

1. Positive regulation / It serves as an activator;
2. Binds to binding site within promoter / specific DNA sequence to help recruit RNA polymerase / increase affinity of RNA polymerase to the promoter;
3. to increase frequency of transcription of *mymA* operon;

(iii) Unlike the wild type bacteria, a mutated *Mycobacterium tuberculosis* has one of the structural genes in the *mymA* operon deleted.

Explain if the gene products coded for by the remaining structural genes in the *mymA* operon of the mutated bacteria are functional.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[3]

1. Gene products coded for by the remaining six structural genes in the mutated bacteria are functional;
2. This is because the expression of the operon results in a polycistronic mRNA with multiple start and stop codons;
3. Hence, each protein is produced independently of the other proteins coded by the same mRNA;

(iv) In another mycobacterium, the operator is deleted.

State the effect of this deletion mutation on the expression of the *mymA* operon.

……………………………………………………………………………………………………

……………………………………………………………………………………………………[1]

1. Operon will be turned on all the time / constitutively expressed;

[Total: 30]

For Examiner's Use

2 Huntington's disease (HD) is a genetic condition that leads to a loss in brain function. The gene involved contains a section of DNA with many repeats of the base sequence CAG. The number of these repeats determines whether or not an allele of this gene will cause Huntington's disease, as illustrated in Fig. 2.1 below.

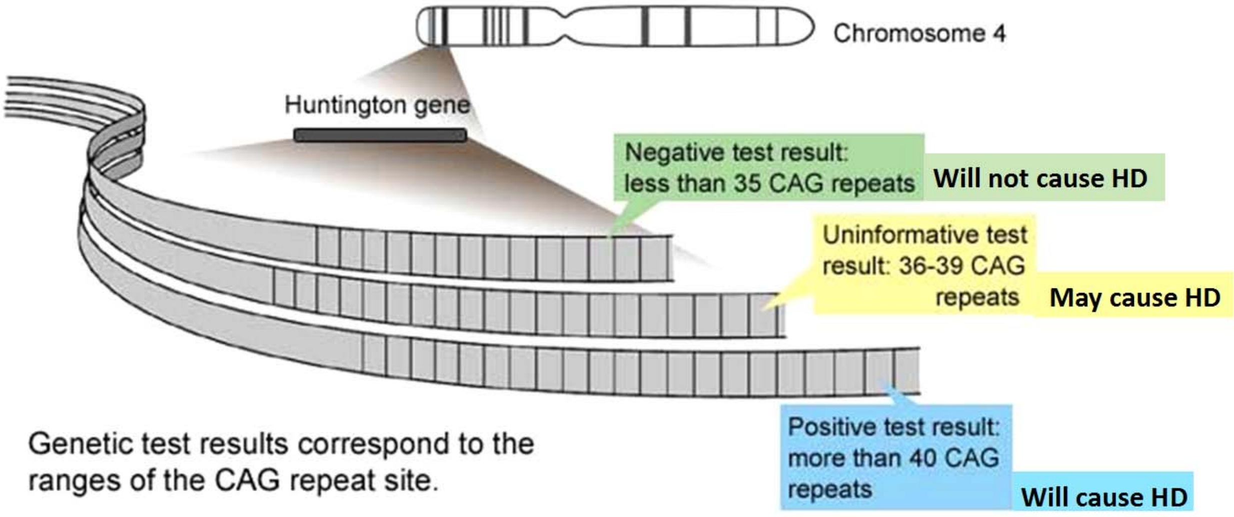


Fig. 2.1

Fig. 2.2 below shows the age at which each patient with HD first developed symptoms and the number of CAG repeats in the allele causing HD disease for each of these patients.

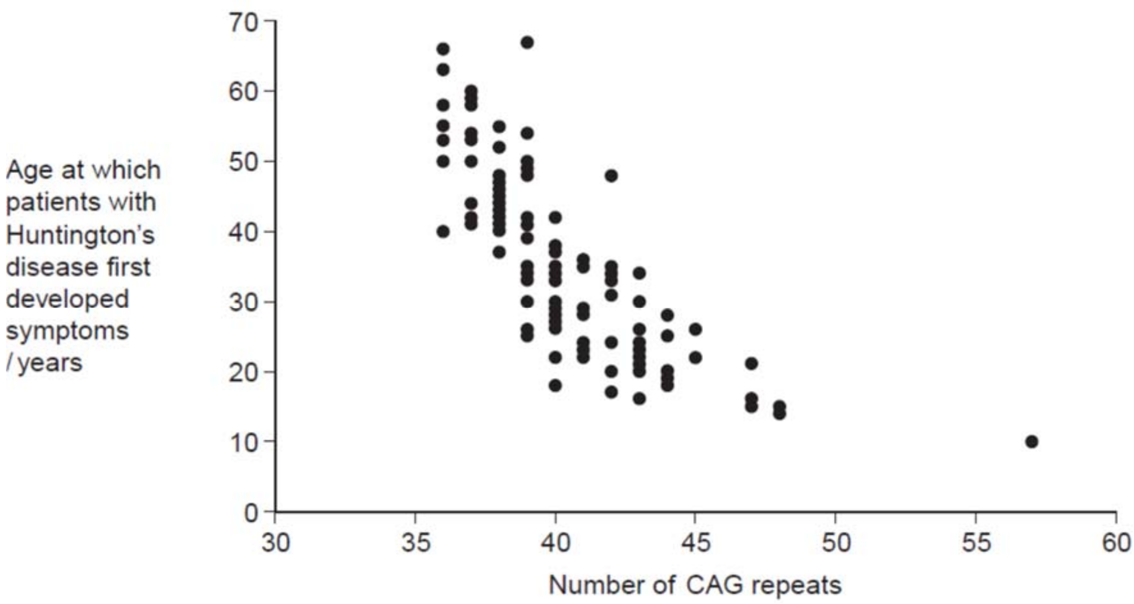


Fig. 2.2

(a) People can be tested to see whether they have an allele for this gene with more than 36 CAG repeats. Some doctors suggest that the results can be used to predict the age at which someone will develop HD.

(i) Use information in Fig. 2.2 to evaluate this suggestion.

....

....

....

....

....

....[3]

1. **As the number of CAG repeats increase, the age at which patients' symptoms develop decreases;**

2. **For example, those with 36 repeats developed symptoms between 40-68 years old, while those with 45 repeats developed symptoms at 20 – 25 years old;**

3. **(Quote values) E.g. <u>Wide data range</u> is observed where with 40 CAG repeats, the age at which patients with HD first developed symptoms ranged from approximately 18 years to 42 years (accept any correct data quoted);**

   OR

   (Quote values) E.g. <u>Overlap</u> in data observed, corresponding to approximately 36 CAP repeats, the age at which patients with HD is recorded as a smear ranging from 40 to 48 years (accept any correct data quoted);

4. **The graph <u>does not linearly correlate</u> the number of CAG repeats to the <u>exact age</u> at which patients with Huntington's disease first developed symptoms / cannot predict <u>exact age</u> for onset of symptoms;**

5. **Graph suggests that <u>other factors</u> beyond number of CAG repeats may be involved in determine age of onset of HD symptoms;**

(ii) Huntington's disease is always fatal. Despite this, the allele is passed on in human populations. Use information in Fig. 2.2 to suggest why.

....

....

....

....[2]

1. **As the <u>onset</u> of symptoms for HD generally <u>appear later</u> in life;**

2. **Thus individuals may have <u>already had children</u> and <u>passed on the allele</u> to their children;**

   OR

3. **Individuals may have <u>already</u> had <u>children</u> and <u>passed on allele</u>;**

4. **<u>Before symptoms occur</u>;**

(b) Scientists took DNA samples from three people, **J**, **K** and **L**. They used the polymerase chain reaction (PCR) to produce many copies of the piece of DNA containing the CAG repeats obtained from each person. They separated the DNA fragments by gel electrophoresis. A radioactively labelled probe was then used to detect the fragments. Fig. 2.3 shows the appearance of part of the gel after an X-ray was taken. The bands show the DNA fragments that contain the CAG repeats.

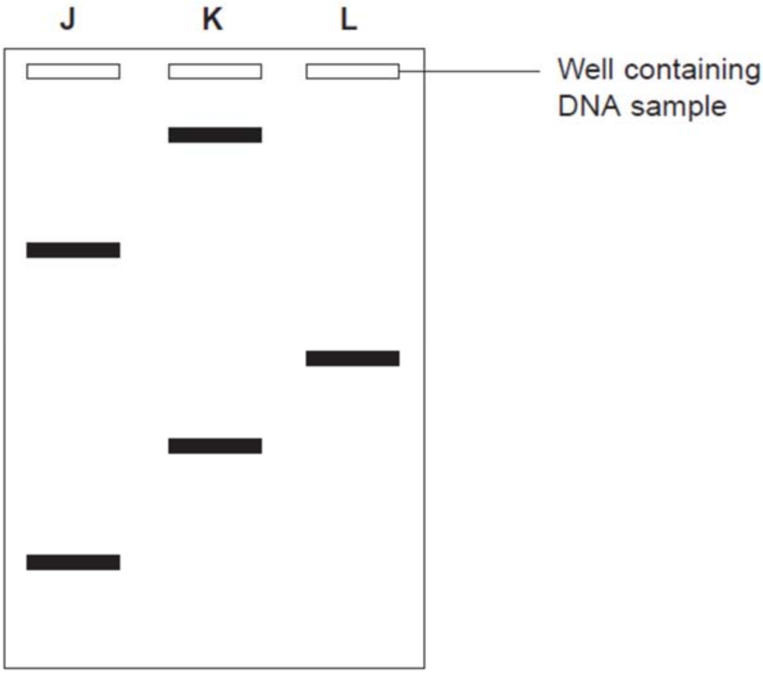


Fig. 2.3

(i) Only one of these people tested positive for Huntington's disease. Explain which person this is.

....

....

....

....[2]

1. **Person K;**
2. **Gel pattern shows that the DNA profile contains a band that had travelled the shortest distance from the wells and hence shows that the DNA contained the most number of CAG repeats;**

(ii) Fig. 2.3 only shows part of the gel. Suggest how the scientists found the number of CAG repeats in the bands on the gel.

....

....[1]

1. **Use DNA ladder having fragments of known length corresponding to known numbers of CAG repeats as a basis of comparison to determine unknown fragment length;**

(iii) Two bands are usually seen for each person tested. Explain why only one band was seen for Person **L**.

………

………

………

………[2]

1. Person L was <u>homozygous</u> and has <u>both</u> alleles with the <u>same number of CAG repeats</u>;
2. Hence <u>same length / size / mass</u> and appears only as one band;

OR

AVP:
3. Person L could have a small fragment that had <u>run off the gel</u>;
4. Occurring if the <u>duration</u> of gel electrophoresis was <u>too long</u>;

[Total: 10]

3 There have been concerns that climate change and global warming could have serious consequences for the survival of the polar bears in the Arctic region. Whilst the actual impact of climate change on polar bear survival is not yet clear, there have been evidence that this could be detrimental to the polar bears in the long term.

Polar bears and their prey have evolved to live in the extreme conditions of the Arctic. Polar bears and seals depend on sea-ice for foraging, resting, and reproduction. The Arctic ecosystem was and continues to be shaped by climate.

There have been a number of climate change scenarios based on modelling of climate trend data in the Arctic. The more extreme models suggest that the Arctic basin will lose its ice within 50 years. Other models have suggested that ice thickness has decreased by 40% during the past 30 years and the average annual ice coverage in the polar region has diminished substantially, with an average annual reduction of over 1 million square kilometers.

Researchers have studied the polar bear population demographics in the Western Hudson Bay area of the United States since 1981. This region is the southern-most extent of the range of the polar bear. They have noticed that the sea ice breakup has been occurring earlier. The earlier breakup has been related to the poorer condition of polar bears and there is a correlation between the earlier breakup and a pattern of warming air temperatures during the spring between 1950 and 1990. It appears that earlier breakup caused by warmer temperatures has resulted in declines in physical and reproductive conditions of polar bears in this area.

Passage adapted from essay by:
Schliebe, S.L. – 'What has been happening to polar bears in recent decades?"
National Oceanic and Atmospheric Administration webite:
https://www.pmel.noaa.gov/arctic-zone/essay_schliebe.html

(a) Explain the term 'global warming'.

....

....[1]

Increase in temperature of earth's surface / atmosphere;

For Examiner's Use

(b) Polar bears (*Ursus maritimus*) are thought to have evolved from brown bears or grizzly bears (*Ursus arctos*). This period coincided with the melting and retreat of the glaciers northward. Fig. 3.1 shows the divergence times of different bears with time in millions of years.

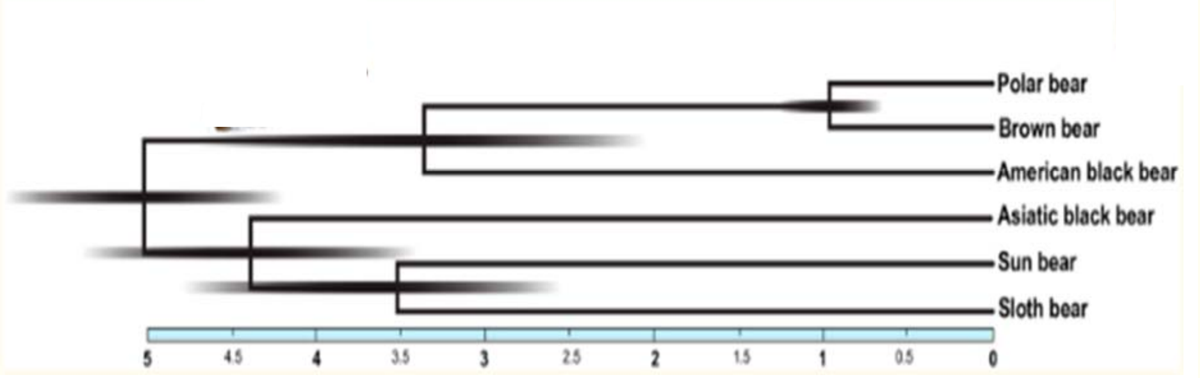


Fig. 3.1

(i) State what is meant by the term, *biological species.*

....

....

....

....[2]

1. **A group of organisms of the same species are capable of <u>interbreeding</u> and producing <u>fertile, viable offspring</u>;**
2. **That is <u>reproductively isolated from other species</u>;**

(ii) Explain how the ancestors of the polar bear could have diverged from the brown bear about 1 million years ago.

....

....

....

....[2]

1. **Ancestors of the polar bear moved <u>northwards</u> with the retreating glaciers;**
2. **They were <u>adapted</u> to the <u>colder climate</u> in the Arctic regions;**

(iii) In recent years, there have been a discovery of hybrids between polar bears and brown bears (grizzly bears) called 'pizzlies' or 'grolars'. Using information from Fig. 3.1, suggest a possible reason why these hybrids are possible.

....

....

....

....[2]

1. **Both grizzlies and polar bears <u>diverged</u> from each other about <u>1 million years ago</u> / shared a common ancestor 1 million years ago;**

2. **Time is too short for them to be too genetically different as to make the hybrid formation impossible / still able to interbreed to form hybrids; (WTTE)**

(iv) Suggest how climate change can result in the polar bears and the brown bears (grizzly bears) interacting and interbreeding to produce these hybrids.

……………………………………………………………………………………………….

………………………………………………………………………………………………..[1]

1. **Warming temperatures allowing brown bears / grizzly bears to move further north, thus coming into possible contact with polar bears; (WTTE)**
2. **Melting sea-ice has forced the polar bears to forage on land thus they come into contact with brown bears; (WTTE)**
3. **Melting sea-ice forcing polar bears to spend less time on ice and less opportunities to hunt seals, thus they move onto land and come into contact with brown bears; (WTTE)**
4. **AVP**

Any one

(c) A student who is doing a project on the effects of climate change on the possible evolution of animals in the Arctic had read about the current trends in climate changes in the Arctic. He thinks that this could lead to the rapid deterioration of the polar bear population in areas like the Western Hudson Bay region.

This student wrote in the draft notes for his written report: "Polar bears may still survive in the face of climate change. For example, if one polar bear had the ability to adapt to warmer temperatures and could hunt effectively on land instead of on sea-ice, this polar bear would evolve into an animal that would be very well suited for the warmer temperatures of the Arctic region in the years to come."

Using your understanding of evolution and natural selection, evaluate the student's claim and justify with appropriate reasons.

…………………………………………………………………………………………..............

…………………………………………………………………………………………..............

…………………………………………………………………………………………..............

…………………………………………………………………………………………..............[2]

1. **The student's claim is incorrect;**
2. **Long-term effects of natural selection are at the level of the population rather than the individual; (Individuals do not change over their lifetimes.) / Microevolution is a measure of changes in allele frequencies in a population over successive generations;**

Reject statements like: "The fate of an individual organism is relatively insignificant in the history of a species." unless there is qualification based on ideas of microevolution and change in allele frequencies in a population.

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the separate writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section (a), (b) etc., as indicated in the question.

4 **(a)** Discuss the importance of ions in Biology. [12]

1. Phosphate ion;
   (a) Forms the **charged** phosphate head of **phospholipids**, allows phospholipid to **interact with aqueous solution** to form a **phospholipid bilayer**;
   (b) Which allows for compartmentalisation within cells / cell membrane serves as a selective barrier for regulating movement of substances in and out the cell;
   (c) Added to **glucose** / phosphorylation of glucose to make glucose **more reactive** for **glycolysis**;
   (d) Added to **relay proteins** during **signal transduction** to **activate / inactivate** the relay proteins / specific example of phosphorylated relay proteins;
   (e) Component of **nucleotides** and therefore allows for the formation of **phosphodiester bonds** / sugar-phosphate backbone in **DNA/RNA**;
   (f) Gives DNA/RNA a **negative charge**, allowing DNA/RNA molecules to be separated via **gel electrophoresis**;

2. Protons;
   (a) **Photolysis of water** during light-dependent stage of photosynthesis, **protons are formed** and will **add on to the high concentration of protons in thylakoid space**;
   (b) **Proton gradient** across thylakoid membrane / cristae, used in the **formation of ATP** by **ATP synthase** during **chemiosmosis**;
   (c) Protons together with electrons reduce NAD^+ / FAD / NADP to **form NADH / $FADH_2$** OR **NADPH** which are used to transport protons and electrons to ETC where electrons are passed on to electron carriers / allows NAD and FAD to serve as **mobile electron carriers** during aerobic **respiration** (A: mention of specific stages in respiration where NADH and/or FADH are formed) OR **reduce GP to G3P** during **Calvin cycle**;
   (d) Gives rise to an **acidic pH** in **lysosomes**, which is the **optimal pH** of **hydrolytic enzymes**;
   (e) Acidic pH within lysosomes allows fusion of influenza viral envelop with endosome membrane, releasing viral genome into host cytoplasm;

3. Zwitterion;
   (a) **Amphoteric** nature / it is able to act both as an acid and base, and therefore serve as **pH buffers** in biochemical reactions;
   (b) If pH becomes **too acidic/alkaline**, **enzymes** can become **denatured**;

4. Iron;
   (a) Component of **haem group** in **haemoglobin**;
   (b) **Binds oxygen** and allows haemoglobin to **transport oxygen**;

5. Ions in saltwater;

Intrusion of saltwater into soil, decreases water potential in soil, leading to excessive loss water from plants;

6. Calcium ions;
   (a) To **neutralise negative charge on DNA** and **phosphate heads of phospholipid bilayer** of cell membrane, allowing for **easier uptake** of DNA into bacterial cell during **transformation** / **no repulsion** between DNA and phospholipids during transformation; ("transformation" marked for once only);

7. Chloride ions;
   Intake is accompanied by water, **priming the cells** for heat shock treatment during **transformation**;

The importance of ions in Biology • A very large number of students wrote very fact-filled essays about ions in Biology but with little or no consideration of their importance. There were several topic areas that were common to the other essay title but with factual content relevant in a different way. • Surprisingly few wrote about ions in biological molecules, and cellular structures, or pH. • Attempts to discuss the roles of ions in haemoglobin were often superficial. • The nitrogen cycle was a popular topic and there were many good accounts of the facts. Few linked the cycle to amino acid or nucleotide synthesis. Some students gave good accounts of the use of fertilisers and the potential problems of eutrophication.

Popular topics: the role of ions in generating and maintaining water potential gradients in the context of cholera, co-transport of glucose, uptake by root hairs and the role of the endodermis in the generation of root pressure. • Cystic fibrosis was used by some students and offered an opportunity for A* content. A discussion of the role of sodium ion gradients in the re-absorption of water in the kidney was also included by many students. • From nerve impulses and synapses, many students moved on to the roles of calcium ions in muscle contraction. Again, not many students attempted to explain importance.

Respiration and photosynthesis were also commonly discussed, with the emphasis on the role of protons. There were good accounts of how proton gradients are generated and then used to produce ATP and/or reduced NADP (in photosynthesis). Once again, students often failed to follow up with the importance of these processes; for example the use of reduced NADP in the synthesis of sugars that can be used for respiration or in the synthesis of other biological molecules. • Poor expression often marred students' accounts – it was quite common to see references to H+ being used/passing down the electron transfer chain. There were some attempts to discuss uses of ATP that involved the transfer of phosphate groups but these tended to be vague.

(b) Insulin and glucagon help to regulate blood glucose concentration. Disruption of this homeostatic mechanism can lead to Type 2 diabetes.

Describe the mode of action of insulin in regulation of blood glucose concentration and explain how disruption of the homeostatic mechanism leads to Type 2 diabetes. [13]

A) General Introduction

1. **Blood glucose regulated by two hormones, insulin and glucagon, which are produced in the islets of Langerhans of the pancreas.**
2. **Beta (β) cells produce insulin, alpha (α) cells produce glucagon.**
3. **Both cell types monitor blood glucose concentration.**

B) Insulin Action (Max 5 – required points 4, 6, 8, at least one from 10-15, 16)

4. **An increase in blood glucose level beyond / above set-point of 90 mg per 100 ml of blood – stimulates insulin secretion; and inhibits glucagon secretion;**
5. **Insulin binds to the receptor tyrosine kinase (RTK) on surface of target cells e.g. muscle cells, liver cells etc**
6. **Initiates ligand-receptor interaction, signal transduction and cellular response;**
7. **Phosphorylation cascade and signal amplification also take place in the target cells; (point 6 marked for once? Point 7 need?)**
8. **Increases the permeability of cell membrane to glucose by increasing the number of glucose carrier proteins / GLUT4 carrier proteins on target cells e.g. muscle cells, liver cells etc;**
9. **Increases rate of glucose uptake by target cells;**
10. **Increase rate of glycogenesis – conversion of glucose to glycogen, this stimulates glycogen storage in liver and muscle;**
11. **Inhibits glycogenolysis – the breakdown of glycogen to glucose in liver and muscle;**
12. **Inhibits gluconeogenesis (glucose synthesis) – the breakdown of fats and proteins and the conversion of non-carbohydrate sources to glucose in liver cells;**
13. **Increases the use of glucose as a substrate for cellular respiration, hence increase rate of glucose oxidation and synthesis of ATP;**
14. **Increases rate of protein synthesis;**
15. **Increases rate of conversion of glucose to fats / rate of fat deposition in adipose tissues;**
16. **When set point decreases to set point / 90 mg glucose per 100 ml blood, secretion of insulin stops;**

C Glucagon Action (max 4 required points – 17, 18, at least one from 21-24, 25)

17. **A decrease in blood glucose level below the set point of 90 mg per 100 ml of blood – stimulates glucagon secretion, inhibits insulin secretion;**
18. **Glucagon binds to the G-protein coupled / linked receptor (GPCR) on surface of target cells e.g. liver cells;**
19. **Initiates ligand-receptor interaction, signal transduction and cellular response; (mark once only – point 6 / point 19)**
20. **Phosphorylation cascade and signal amplification also take place in the target cells; (mark once only – point 7 / point 20)**
21. **Increases rate of glycogenolysis – the breakdown of glycogen to glucose in liver; (Note that glucagon has no effect on muscle glycogen)**
22. **Increases rate of gluconeogenesis (glucose synthesis) – the breakdown of fats and proteins and the conversion of non-carbohydrate sources to glucose in liver cells. (e.g. Increases rate of conversion of lactic acid to glucose.);**
23. **Increases rate of breakdown of lipids into fatty acids (lipolysis);**
24. **Inhibits glycogenesis;**

25. When set point increases to 90 mg glucose per 100 ml blood, secretion of glucagon stops;

Type 2 Diabetes – Loss of Homeostatic Mechanism (max 3 – points 26, 27, 29/30 required)

26. Insulin normally produced but does not bind to RTK (insulin resistance);
27. Vesicles containing GLUT4 carrier proteins do not move to cell membrane of target cells and so glucose is not taken up normally;
28. Glycogen not stored in liver and muscle cells;
29. Blood glucose remains high – above set point of 90 mg per 100 ml of blood;
30. Blood glucose not regulated – may be too high (hyperglycaemia) after a meal or drops too low (hypoglycaemia) as not enough stored glycogen for breakdown to release glucose;

QWC: Clear and systematic description of modes of action both insulin and glucagon covering both physiological and molecular events AND clear link to loss of homeostatic control leading Type 2 diabetes.

5 **(a)** Carbon dioxide may affect organisms directly or indirectly. Describe and explain these effects. [12]

DIRECT effects of CO_2 on organisms (**max 3**)

1. Carbon dioxide is used for **carbon fixation** during **light-independent stage** of **photosynthesis** / **Calvin cycle**;
2. Whereby carbon fixed is used to produce **glyceraldehyde-3-phosphate** / **glucose**;
3. At **higher [CO_2]**, rate of carbon fixation will decrease and **rate of formation of glucose** will **increase** (ORA);
4. This will lead to **increased growth** of plants as glucose is used to form **cellulose** / used to **produce more ATP** for growth;
5. This can **increase yield** of **crop plants**;

6. Respiration, photosynthesis, activity giving rise to short-term fluctuations and long-term change (direct effects of increasing CO_2) ??

INDIRECT effects of (increasing) CO_2 on organisms (**max 8**)

7. Carbon dioxide is a **greenhouse gas** and hence **traps heat** in Earth's atmosphere and results in **global warming**; (must have)

Effect of global warming on physio:

8. This can result in **increased rate** of **transpiration** / **water loss**, resulting in **wilting** of plants;
9. Plant may adapt via **reduced stomatal opening** to **reduce water loss** through stomata / transpiration which will then affect **photosynthesis / respiration** due to **reduced uptake of CO_2 / O_2**;
10. This will lead to **decreased growth** of plants;

11. Increased temperatures may **increase rate** of **respiration / metabolic reactions**, leading to **increased growth** of organisms;
12. e.g. increased growth of some **crops**, e.g. those grown in **temperate countries**
    OR
    e.g. increased growth and reproduction of **bacteria** / increased growth and reproduction of **insects and pests**, leading to **increased incidence of disease**;
13. This is because **kinetic energy** of enzymes and substrates will **increase**, leading to **increased rate of effective collisions** between enzymes and substrates and **increased rate of formation of enzyme-substrate complexes**;

14. However, if temperature **exceeds optimal temperature** of organisms, enzymes will **denature**;
15. Increased temperatures can also lead to **heat stress** in animals, leading to **decreased appetite** / **increase in death** of livestock;

16. Increased temperatures can also cause organisms to **move / migrate** to **higher latitudes** / **higher elevations**;

17. For organisms that are **unable to adapt / move to a new favourable location**, **extinction** of species may occur;

18. **QWC** – response is clear in terms of direct and indirect effects of carbon dioxide;

(b) Discuss the control of processes in cells and the importance of these controls. [13]

Control of cell division (**max 4**)

1. Cell cycle **checkpoints** are regulated by **tumour suppressor gene and proto-oncogene products**;
2. Proto-oncogene products stimulate **normal cell growth and proliferation / cell division**;
3. Tumour suppressor gene products are involved in **inhibiting cell growth and division** / **activate cell cycle arrest**, DNA repair and/or apoptosis;
4. **G1 checkpoint** assesses if all organelles have been duplicated **before cell cycle is allowed to progress** into mitosis / meiosis;
   OR
   G2 checkpoint assesses **if DNA replication is completed** and cell size is adequate, **before cell cycle is allowed to progress** into mitosis / meiosis;
   OR
   Metaphase checkpoint assesses if **spindle fibres** are **attached** to **all chromosomes** before cell is allowed to enter **anaphase**;
5. **Loss-of-function mutation** in tumour suppressor genes and **gain-of-function mutation** in proto-oncogenes can lead to **dysregulation of cell cycle checkpoints**;
6. **Dysregulation of cell cycle checkpoints** can result in **uncontrolled cell division**, leading to **tumour formation**; (must have)

Control of gene expression in cells (**max 4**)

7. In eukaryotes, **gene expression** is **upregulated** / turned "on" via **histone acetylation** / **chromatin remodelling complexes**;
8. **Chromatin** is **decondensed**, allowing **RNA polymerase** and **general transcription factors** to **bind** to **promoter** to form **transcriptional initiation complex**, allowing transcription to take place;
9. In eukaryotes, **gene expression** is **down-regulated** / repressed via **DNA methylation** / repressive **chromatin remodelling complexes**;
10. **Chromatin** is **condensed**, preventing **RNA polymerase** and **general transcription factors** to **bind** to **promoter**, and **transcriptional initiation complex** cannot form, preventing transcription;
11. This allows for **differential gene expression** when cells undergo **differentiation** / cells become specialised; (must have)
12. In prokaryotes, all genes are under the control of the same promoter in an operon. Operon is **switched "off"** when **repressor binds** to **operator**; (ORA)
13. **Mutation** in repressor gene will result in **constitutive expression** of operon, **wasting cell resources**; (must have)

Control of blood glucose concentration (**max 4**)

14. **Beta cells** of the **islets of Langerhans of pancreas** only **secretes insulin** when blood glucose concentration goes **above 90mg/dL**;
15. **Alpha cells** of the **islets of Langerhans of pancreas** only **secretes glucagon** when blood glucose concentration drops **below 90mg/dL**;
16. In response to **insulin**, processes such as **glycogenesis** and **transport of glucose transporters to cell membrane** will be **upregulated** in liver cells / muscle cells to help decrease blood glucose concentration;
17. In response to **glucagon**, processes **gluconeogenesis** and **glycogenolysis** will be **down-regulated** in liver cells to help increase blood glucose concentration;
18. Disruption of such homeostatic controls will lead to **diabetes**; (must have)

19. QWC – at least three different areas of control in cells

Blood glucose regulation – pancreas, islets of Langerhans, Insulin, glucagon, liver,

Cell signaling – signal reception and signal termination

Transport – channels + carriers – control size and shape & properties of mlc to be transported

Enzyme-catalysed reactions – specificity, feedback inhibition, allosteric activation / inhibition – PFK

Photosynthesis – only when have light then does PS take place / temperature can also be a limiting factor

Respiration – temperature as limiting factor

Inheritance – inactivation of X chromosome

The control of processes in cells and the importance of these controls • Some weaker essays included long, rambling discussions of the importance of control with little relevant detail to illustrate the points being discussed. • Some students ignored 'in cells' and wrote at length about systems in organisms: for example homeostatic control systems in mammals. • Popular examples used: cholera, cystic fibrosis, ion movement into roots related to control of water uptake, enzymes and factors controlling their activity, transcriptional factors, proto-oncogenes and tumour suppressor genes – possible triggers by questions in the paper, control of DNA replication and cell division. • Some students showed poor recollection of the control of blood glucose concentration and many failed to discuss its importance. There was considerable confusion about the different cellular mechanisms involved in the production of oestrogen, insulin, glucagon and adrenaline.

There were some potentially A* accounts of ADH in control of water uptake in the kidney. • Some of the better attempts to discuss control were associated with the accounts of the roles of IAA in controlling tropisms. • Accounts of the nervous system were very common. The best included consideration of the importance of unidirectional transfer of information across synapses, the control of the propagation of a nerve impulse and the importance of the refractory period. Many accounts were heavy on facts but with limited consideration of control. • There were many accounts of ATP, respiration and photosynthesis but these were frequently little more than statements of fact, with few references to control.